

The big picture

Guiding question(s)

What happens when protons are transferred?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

We have been using acids and bases in everyday life for thousands of years. The Ancient Greeks put forward the first ideas on acidity when they defined sour-tasting substances as acids. The Egyptians also had a similar definition based on taste. Both the Greeks and the Egyptians discovered that one substance in particular was very sour. This substance became known as vinegar (a solution of ethanoic acid). Consequently, they developed a new category of substances, termed acids, that included all things that were sour, or sharp tasting, including lemon juice. The Latin word *acidus* means sour.

The Greeks also became familiar with a further, different category of substance – bases. They had found three slippery substances as residue after certain materials were burned. These three substances or residues that had a soapy feel to the skin were: soda, potash, and lime.

The word alkali was developed from the Arabic word *al-qaliy*, meaning ‘the ashes’, and was then used when talking about substances that felt slippery to the touch.

The term alkali is now used more specifically to describe a base that dissolves in water.

Watch **Video 1** to find out more about acids and alkalis.

What Are Acids & Bases? | Chemistry Basics



Video 1. Acids and bases.

Concept

Reactivity is one of the two main concepts in the DP chemistry course, with the other being structure. We have previously seen that structure in chemistry is concerned with the way in which the particles of a substance are arranged. Reactivity is concerned with how and why substances interact with each other. In this subtopic, the reactions of acids with bases and other substances will be explored.

Prior learning

Before you study this subtopic make sure that you understand the following:

- Writing balanced equations (see [section R2.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/>)).
- Equilibrium (see [sections R2.3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-equilibrium-constant-kc-and-magnitude-of-the-kc-id-46067/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-equilibrium-constant-kc-and-magnitude-of-the-kc-id-46067/>), [R2.3.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-reaction-quotient-q-hl-id-46068/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-reaction-quotient-q-hl-id-46068/>), [R2.3.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/>)).
- Organic structures (see [sections S3.2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/organic-structures-id-43484/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/organic-structures-id-43484/>)).

[753301/book/types-of-formula-id-43467/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/types-of-formula-id-43467/)), [S3.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/) 
(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/>)).

Practical skills

Once you have completed this subtopic, apply your knowledge of acid—base reactions by going to [Practical 8a: Acid—base titration with a chemical indicator](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-titration-with-a-chemical-indicator-8a-id-46724/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-titration-with-a-chemical-indicator-8a-id-46724/>) and [Practical 8b: Acid—base titration with a pH sensor](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-titration-with-a-ph-sensor-8b-id-46725/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-titration-with-a-ph-sensor-8b-id-46725/>).